

THREE RIGIDITIES OF TOURNAMENTS: SPECTRAL, TOPOLOGICAL, AND ARITHMETIC

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ABSTRACT. A *tournament* is a complete directed graph. Let $H(T)$ count directed Hamiltonian paths. We report three previously unknown rigidities. **Arithmetic:** the odd integers 7 and 21 are permanently absent from $\{H(T)\}$ over all tournaments on all n ; and a signed Hamiltonian permanent $S(T)$ satisfies $S(T) \equiv c_n \pmod{2^{n-1}}$ for a constant c_n independent of T . **Spectral:** $H(T)$ has a complete closed-form Walsh–Fourier decomposition; the nonzero coefficients are supported exactly on unions of edge-disjoint even-length paths in K_n , and this simultaneously yields an elementary proof of the Odd-Cycle Collection Formula $H(T) = I(\Omega(T), 2)$ bypassing P -partition theory, and a new link between the statistics of random tournaments, the Cayley transform, and Delannoy lattice paths. **Topological:** in GLMY path homology, $\beta_2(T) = 0$ for every tournament — combinatorial completeness forbids the twin-vertex configurations that force $\beta_2 > 0$ in general directed graphs — while Paley tournaments exhibit a Betti profile matching the cohomology of a Heisenberg Lie algebra.

1. BACKGROUND AND THE ODD-CYCLE COLLECTION FORMULA

A **tournament** T on $[n] = \{1, \dots, n\}$ is a complete directed graph; $H(T)$ denotes the number of directed Hamiltonian paths. By Rédei’s theorem (1934), $H(T)$ is always odd. The **conflict graph** $\Omega(T)$ has one vertex per directed odd cycle of T , with edges between cycles sharing a vertex.

Theorem 1.1 (Odd-Cycle Collection Formula (OCF)). *For every tournament T , $H(T) = I(\Omega(T), 2)$, where $I(G, x) = \sum_{k \geq 0} \alpha_k(G) x^k$ is the independence polynomial.*

The formula was discovered computationally here, verified exhaustively for $n \leq 6$ (33,864 labeled tournaments), and sampled for $n = 7$ –10 with zero failures. An independent proof using P -partitions appears in [1]. Section 3 gives a new, elementary Walsh–Fourier proof. The expansion $H = 1 + 2\alpha_1 + 4\alpha_2 + \dots$ recovers Rédei’s theorem and identifies $H = 1$ with transitivity.

2. ARITHMETIC RIGIDITY

The H -spectrum gap problem. Which odd integers are *permanently forbidden* — absent from $\{H(T)\}$ over all tournaments on all n ?

Theorem 2.1 (Gap Theorem). *$H(T) \neq 7$ and $H(T) \neq 21$ for every tournament T , regardless of n . These are the only permanently forbidden values below 200; $H = 63$ is achieved at $n = 8$.*

Proof sketch for $H \neq 7$. By Theorem 1.1, $H = 7$ requires $\Omega(T) = K_3$ (exactly three odd cycles forming a complete conflict graph). One shows: for $n \leq 7$, no tournament has exactly three pairwise-conflicting cycles (exhaustive at $n \leq 7$, using the fact that at $n = 5$ three mutually-conflicting 3-cycles always generate a 5-cycle). For $n \geq 8$, the three cycles span at most 7 vertices, so the induced subtournament on those vertices reduces to the $n \leq 7$ case. $\square$

The proof of $H \neq 21$ is deeper, proceeding by a case analysis of possible conflict graph structures realizing $I(G, 2) = 21$: the disconnected case is ruled out using $H \neq 7$; the path P_4 is ruled out by a forced-cycle lemma; the remaining cases are closed inductively via a “poisoning-DAG” argument

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All numbered results carry complete proofs; Theorem 4.2 is computationally established for $p = 7, 11$ with an algebraic proof in progress. Preprint in preparation.

showing every sufficiently large cycle-rich tournament either has a good deletion vertex (reducing to a smaller case) or has $H \geq 25 > 21$.

Connection to k -nacci matrices. Let M_k be the companion matrix of the k -th order Fibonacci recurrence $a_n = a_{n-1} + \dots + a_{n-k}$. Newton's identity gives $\text{Tr}(M_k^3) = 7$ for all $k \geq 3$, since the first three elementary symmetric polynomials of k -nacci roots are identical for all such k . For $k = 3$ (tribonacci), $\text{Tr}(M_3^5) = 21 = 3 \times 7$.

Theorem 2.2 (k -nacci Identity). $\text{Tr}(M_k^n) = 2^n - 1$ for all $1 \leq n \leq k$. At $n = k + 1$ the identity breaks by exactly $k + 1$: $\text{Tr}(M_k^{k+1}) = 2^{k+1} - k - 2$.

The forbidden values $\{7, 21\}$ are thus the first two k -nacci traces that lie strictly below their corresponding power-of-two targets.

The signed Hamiltonian permanent. Let $B = 2A - J$ (± 1 signed adjacency matrix) and $S(T) = \sum_P \prod_{i=1}^{n-1} B[P_i, P_{i+1}]$ over all $n!$ vertex orderings. One has $S(T) = 0$ for even n by a reversal-pairing argument.

Theorem 2.3 (Universal Congruences). (i) $S(T) \bmod 2^{n-1}$ depends only on n .

(ii) $S(T)$ is fully independent of T modulo 2^{n-1} if and only if $s_2(n-3) \leq 1$, where s_2 is the binary digit sum. The universal odd values of n are $3, 5, 7, 11, 19, 35, 67, \dots$ (i.e., $n = 3 + 2^k$ for $k \geq 0$, together with $n = 3$).

(iii) (Master Identity.) For any k pairwise non-adjacent positions in a Hamiltonian path, $D_S = n!/2^k$ holds pointwise for all tournaments, where D_S sums weighted edge products over all $(2k)!$ orderings of $2k$ vertices at those positions.

At the first non-universal value $n = 9$, the residue $S(T) \bmod 256$ depends precisely on the parity of the 3-cycle count $t_3(T)$.

3. SPECTRAL RIGIDITY: WALSH-FOURIER THEORY

Encode T as $t \in \{0, 1\}^m$, $m = \binom{n}{2}$. Every tournament invariant expands in the Walsh-Hadamard basis $\{(-1)^{t \cdot \chi_S}\}_{S \subseteq [m]}$.

Theorem 3.1 (Walsh Spectrum of H). $\hat{H}[S] \neq 0$ if and only if S is a union of edge-disjoint even-length paths in K_n . For such S with $|S| = 2k$ and r path components,

$$\hat{H}[S] = \varepsilon_S \cdot \frac{2^r \cdot (n - 2k)!}{2^{n-1}}, \quad \varepsilon_S = \pm 1 \text{ (path-orientation parity)}.$$

Proof. For a path-union monomial S , the inner product $\hat{H}[S] = 2^{-m} \sum_T H(T) (-1)^{t \cdot \chi_S}$ counts Hamiltonian paths weighted by arc consistency with S . Each component of S must appear as a contiguous block in any contributing path; even-length components contribute sign $+1$ regardless of traversal direction (two orientations, each with sign product $(-1)^{2a} = 1$), giving $2^r (n - 2k)!$ oriented placements. For non-path-union S the sum cancels by reversal symmetry. Setting $\hat{H}[S] = \hat{I}_\Omega[S]$ via an independent generating-function calculation proves Theorem 1.1. $\square$

The formula compresses $H(T)$ at $n = 7$ from 2^{21} configurations to ≈ 20 independent Walsh amplitudes (10^5 -fold reduction).

Corollary 3.2 (Transfer Matrix Symmetry). Let $M[a, b] = \#\{\text{directed Hamiltonian paths } a \rightarrow b\}$. Then $M[a, b] = M[b, a]$ for all tournaments and all vertices a, b .

This is non-obvious: the matrix counting directed-path endpoints is symmetric despite the underlying directed structure. It follows from an explicit closed-form Walsh expansion of $M[a, b]$, manifestly symmetric in a, b .

The Cayley–Delannoy connection. Let $g_k(m)$ denote the Taylor coefficients of the Cayley transform: $Q(x)^m = \left(\frac{1+x}{1-x}\right)^m = 1 + 2 \sum_{k \geq 1} g_k(m) x^k$.

Theorem 3.3 (Delannoy Identity).

$$k \cdot g_k(m) = \sum_{j=1}^{\min(k,m)} j \binom{k}{j} \binom{m}{j} 2^{j-1} = \#\{\text{diagonal steps in all Delannoy paths } (0,0) \rightarrow (k,m)\}.$$

The squared coefficient of variation of $H(T)$ over uniformly random tournaments satisfies

$$\text{CV}^2(H) = \sum_{k \geq 1} \frac{2g_k(n-2k)}{\binom{n}{2k}} = \frac{2}{n} + \frac{0}{n^2} - \frac{14}{3n^3} + O(n^{-4}).$$

The sequence $W(n) = n!(1 + \text{CV}^2(H)) = 1, 2, 8, 32, 158, 928, 6350, 49752, \dots$ is not in OEIS.

The exact vanishing of the n^{-2} coefficient follows from $g_2 = g_1^2$. The Cayley transform $Q(x) = (1+x)/(1-x)$ – the bilinear/Tustin transform of digital signal processing – governs the statistics of random tournaments through weighted Delannoy path enumeration; no prior literature records this. Its logarithm $2 \operatorname{arctanh}(x)$ is simultaneously the Poincaré disk geodesic and the Fisher–Rao distance on Bernoulli distributions: *the Fourier energy of a random tournament equals the statistical distance between its arc-distribution and the fair coin.*

4. TOPOLOGICAL RIGIDITY: GLMY PATH HOMOLOGY

GLMY path homology [2] assigns Betti numbers $\beta_p(T)$ via an “allowed paths” chain complex. For general oriented graphs $\beta_2 > 0$ occurs freely (e.g. among 5-vertex oriented graphs); for tournaments it is structurally forbidden.

Theorem 4.1 ($\beta_2 = 0$). *For every tournament T on every number of vertices, $\beta_2(T) = 0$.*

Proof sketch. Induction via the long exact sequence of the pair $(T, T \setminus v)$. By induction $H_2(T \setminus v) = 0$, so $H_2(T)$ injects into the relative term, which vanishes iff $\beta_1(T \setminus v) \leq \beta_1(T)$; it suffices to find one such *good vertex* in every tournament. Four cases cover all possibilities: if $\beta_1(T) = 1$, every vertex is good; if T is not strongly connected, β_1 is 0 throughout and every vertex is good; if T has a cut vertex, that vertex is good; for 2-connected T at $n \geq 6$, a Key Lemma (no cycle can be completely isolated from dominated cycles in a 2-connected tournament) produces at least $n - 5$ good vertices; and $n \leq 5$ is checked exhaustively. $\square$

Structural explanation. Every oriented graph with $\beta_2 > 0$ contains *twin vertices* (two vertices with identical neighborhoods). Tournament completeness forbids twins: if u, v shared the same in-neighborhood, the arc between u and v would simultaneously be required in both directions.

Further proved results: $\beta_1(T) \in \{0, 1\}$ (no tournament has more than one independent 1-cycle); $\beta_1(T) \cdot \beta_3(T) = 0$ (no tournament simultaneously carries 1-cycles and 3-cycles); $\operatorname{rank}(\partial_2) = \binom{n}{2} - n + 1 - \beta_1(T)$ universally. At $n = 8$, $\beta_3 = 2$ first occurs (0.08% of tournaments) and the seesaw $\beta_3 \cdot \beta_4 = 0$ fails, opening new structure.

Theorem 4.2 (Paley Betti Formula; computationally established for $p = 7, 11$). *For the Paley tournament T_p ($p \equiv 3 \pmod{4}$ prime), $a \rightarrow b$ iff $b - a$ is a nonzero quadratic residue mod p , $m = (p - 1)/2$:*

$$\beta_m = \frac{m(m-3)}{2}, \quad \beta_{m+1} = \binom{m+1}{2}, \quad \beta_d = 0 \ (d \neq 0, m, m+1), \quad \chi(T_p) = p.$$

The formula $\beta_m = m(m-3)/2$ equals exactly the second Betti number of the m -dimensional Heisenberg Lie algebra (Santharoubane [3]); this identification appears to be new. Key computations: $H(T_{11}) = 95,095$ and $H(T_{11})/|\operatorname{Aut}(T_{11})| = 1729$ (Hardy–Ramanujan taxicab number).

Theorem 4.3 (Constant Symbol Matrix). *For any circulant tournament, the Tang–Yau symbol matrix $M_m(t)$ [4] is constant in t : all n eigenspaces of the $\mathbb{Z}_n$ -action on the chain complex have identical Ω -dimension vectors.*

Proof. In Tang–Yau’s framework, t -powers arise only from face-0 maps that land outside the allowed-path space A_{m-1} . For a circulant *tournament* (antisymmetric connection set $S \cap (-S) = \emptyset$), one shows: the tail of any allowed m -path is always an allowed $(m-1)$ -path (Key Lemma, proved by comparing partial sums modulo n). Hence face-0 maps never leave A_{m-1} , so $M_m(t) = M_m(1)$ is constant. $\square$

In [4] the symbol matrix varies with t for general circulant digraphs; Theorem 4.3 is specific to tournaments and strictly sharpens their stability theorem. Combined with small-prime Gaussian elimination (uint8 mod $p < 256$, $8\times$ memory reduction, certified at two primes), the complete Betti profile of T_{11} was computed; T_{19} is within reach.

5. OPEN PROBLEMS

- (1) **Gap density.** Are 7 and 21 the only permanently forbidden H -values? Does the achievable density approach 1?
- (2) **Paley Betti formula.** Prove Theorem 4.2 for all Paley primes; the natural approach uses the Heisenberg isomorphism and the representation theory of $\mathbb{Z}_p \rtimes \mathbb{Z}_m$.
- (3) **Paley maximization.** Prove $H(T_p) = \max_{|V|=p} H(T)$ for all $p \equiv 3 \pmod{4}$. Verified: $p = 3, 7, 11$; matches OEIS A038375.
- (4) $\beta_3 = 2$ **structure.** Characterize the 0.08% of 8-vertex tournaments with $\beta_3 = 2$. Is $\sup_T \beta_k(T)$ unbounded in n ?
- (5) **Universality mechanism.** Give a direct 2-adic or representation-theoretic explanation for the universality criterion $s_2(n-3) \leq 1$ in Theorem 2.3(ii).

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